

Anish Rao

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EDUCATIONAL EXPERIENCES

Department of Chemistry <i>Assistant Professor</i> BITS Pilani, Rajasthan, India	December 2025 - present
University of Fribourg <i>Postdoctoral Researcher</i> Supervisor: Prof. Guillermo P. Acuna	January 2025 - November 2025
Centro de Física de Materiales (CFM) <i>Juan de la Cierva Fellow</i> Supervisor: Prof. Marek Grzelczak	January 2023 - December 2024
CIC biomaGUNE <i>Postdoctoral Researcher</i> Supervisor: Prof. Luis M. Liz-Marzán	September 2020 - November 2022
Indian Institute of Science Education and Research (IISER) Pune <i>Integrated PhD Chemistry (MS + PhD)</i> PhD Thesis Supervisor: Prof. Pramod Pillai	2013 - 2020 <i>Overall CGPA: 8.3/10</i>
Ramjas College, University of Delhi <i>B.Sc. (H) Chemistry</i>	2010 - 2013 <i>Percentage: 73.4%</i>

PROFESSIONAL EXPERIENCE

POSTDOCTORAL RESEARCHER Advisor: Prof. Guillermo P. Acuna - Surface engineering and self-assembly of high-index silicon NPs (SiNPs) for DNA programmed crystallization. - DNA origami - directed assembly of SiNP dimers for chiral sensing.	<i>January 2025 - November 2025</i> <i>Univ. of Fribourg</i>
JUAN DE LA CIERVA FELLOW Advisor: Prof. Marek Grzelczak - Utilizing Bayesian optimization, machine learning, and automation to improve mechanistic understanding during the growth of gold nanorods (AuNRs). - Designing and utilizing DNA-coated gold nanoparticles (AuNPs - <i>spherical nucleic acids</i>) for obtaining hydrodynamic oscillations. - Developing a self-driving platform for closed-loop gold bipyramid (AuBP) synthesis.	<i>January 2023 - December 2024</i> <i>CFM</i>
POSTDOCTORAL ASSOCIATE Advisor: Prof. Luis M. Liz-Marzán - Utilizing nanoscale forces to form AuNR-quantum dot (QD) assemblies. - Here, I utilized suitably chosen AuNRs to enhance the emission of QDs. In addition for their use in photonic integrated circuits, such systems have promising applications for bioimaging probes. - Developing self-assembly protocols for integrating bottom-up and top-down fabrication strategies. - Studying the effect of nanoparticle volumes (varying absorption and scattering contributions) on	<i>September 2020 - November 2022</i> <i>CIC biomaGUNE</i>

the photo-thermal heating abilities of AuNRs. Here, I prepared 8 different AuNR samples with the same aspect ratios, leading to the same extinction spectrum, but different volumes. We then studied the photo-thermal heating abilities of the AuNRs.

GRADUATE RESEARCH ASSISTANT

2015 - 2020

Advisor: Prof. Pramod Pillai

IISER Pune

- Here, the main idea of my thesis was to demonstrate that suitable control over interparticle interactions can:-
 - **improve** existing properties of AuNPs - preparing reusable sensors.
 - **impart** inherently absent properties to AuNPs - preparing selective sensors without the use of analyte-specific reagent.
 - help in **mimicking** life-like behaviour with NP building blocks - preparing transient self-assemblies.
- First, I regulated different interparticle interactions to impart stability to AuNP aggregates (*an inherently unstable entity*).
- Used these stable AuNP aggregates to trap and scavenge heavy metal ions selectively and reversibly!
- Next, I demonstrated the **emergence of selectivity** towards Pb^{2+} ions using systems composed of inherently nonselective AuNPs.
- Here, I used *preferential breaking of interparticle interactions* to design a highly versatile and tuneable 'turn-on' identification assay.
- Finally, I could introduce temporal fluctuations in interparticle interactions to show the formation of transiently stable nanoparticle precipitates (*Dynamic Self-Assembly*).
- In addition to working on self-assembly, and sensing projects (as a part of my thesis), I have mentored undergraduate students in the areas of AuNP-catalyzed reactions as well. Here, one of our most remarkable studies showed that under suitable experimental conditions, *a weaker reducing agent could outperform a stronger reducing agent!*

PROJECTS DURING MASTERS

December 2014 - April 2015

Advisor: Prof. Pramod Pillai

Masters Project

- Developed multi-functional nanoparticles with superior stability compared to conventional negatively charged AuNPs.
- Developed place-exchange strategies to synthesize and functionalize AuNPs with two oppositely charged ligands on the surface of same nanoparticle, with varying proportions.
- Studied factors affecting the proportion of ligands on the surface of AuNPs.

LAB ROTATIONS

December 2013 - September 2014

- **Advisor: Prof. Sujit K. Ghosh** Learned about the design, synthesis, and characterization of metal-organic frameworks (MOFs).
- **Advisor: Prof. Anirban Hazra** Learned quantum mechanical calculations (Hartree Fock method), and studied the effect of theory as well as basis set use on modelling Haber-Bosch process.
- **Advisor: Prof. Partha Harza** Studied the photophysical properties of an anticancer drug (berberine) in micelles.
- **Advisor: Prof. Harinath Chakrapani** Learned about the design and synthesis of prodrugs.

I have had an enriching and multidisciplinary research experience of working in various research fields, across different research labs.

PUBLICATIONS

For a regularly updated list of publications, check out my [Google Scholar](#).

1. **Rao, A.,** Roy, S., Unnikrishnan, M., Bhosale, S. S., Devatha, G., and Pillai, P. P., Regulation of Interparticle Forces Reveals Controlled Aggregation in Charged Nanoparticles *Chem. Mater.*, **2016**,

28, 2348 – 2355.

2. Devatha, G., Roy, S., **Rao, A.**, Mallick, A., Basu, S., Pillai, P. P., Electrostatically Driven Resonance Energy Transfer in “Cationic” Biocompatible Indium Phosphide Quantum Dots *Chem. Sci.*, **2017**, *8*, 3879 – 3884.
3. Roy, S., **Rao, A.**, Devatha, G., Pillai, P. P., Revealing the Role of Electrostatics in Gold – Nanoparticle – Catalyzed Reduction of Charged Substrates *ACS Catal.*, **2017**, *7*, 7141 – 7145.
4. Xavier, J. A. M., Devatha, G., Roy, S., **Rao, A.**, Pillai, P. P., Electrostatically Regulated Photoinduced Electron Transfer in “Cationic” Eco-friendly CuInS₂/ZnS Quantum Dots in Water *J. Mater. Chem. A*, **2018**, *6*, 22248 – 22255.
5. Roy, S., Roy, S., **Rao, A.**, Devatha, G., Pillai P. P., Precise Nanoparticle – Reactant Interaction Outplays Ligand Poisoning in Visible–Light Photocatalysis *Chem. Mater.*, **2018**, *30*, 8415 – 8419.
6. Chakraborty, I.; Roy, S.; Devatha, G.; **Rao, A.**; Pillai, P. P., InP/ZnS Quantum Dots as Efficient Visible-Light Photocatalysts for Redox and Carbon-Carbon Coupling Reaction *Chem. Mater.*, **2019**, *31*, 2258 – 2262.
7. Devatha, G., **Rao, A.**, Roy, S. Pillai, P. P., Förster Resonance Energy Transfer Regulated Multicolor Photopatterning from Single Quantum Dot Nanohybrid Films *ACS Energy Lett.*, **2019**, *4*, 1710 – 1716.
8. **Rao, A.**, Kumar, G. S., Roy, S., Ajesh, T. R., Devatha, G., Pillai, P. P., Turn-On Selectivity in Inherently Nonselective Gold Nanoparticles for Pb²⁺ Detection by Preferential Breaking of Interparticle Interactions *ACS Appl. Nano Mater.*, **2019**, *2*, 5625 – 5633.
9. Roy, S., Jain, V., Kashyap, R. K., **Rao, A.**, Pillai, P. P.*, Electrostatically Driven Multielectron Transfer for the Photocatalytic Regeneration of Nicotinamide Cofactor *ACS Catal.*, **2020**, *10*, 5522 – 5528.
10. Devatha, G., Roy, P., **Rao, A.**, Roy, S., Pillai, P. P.*, Multicolor Luminescent Patterning via Photoregulation of Electron and Energy Transfer Processes in Quantum Dots *J. Phys. Chem. Lett.*, **2020**, *11*, 4099 – 4106.
11. Roy, P., Devatha, G., Roy, S., **Rao, A.**, Pillai, P. P.*, Electrostatically Driven Resonance Energy Transfer in an All-Quantum Dot Based Donor–Acceptor System *J. Phys. Chem. Lett.*, **2020**, *11*, 5354 – 5360.
12. Shirin, S., Roy, S., **Rao, A.***, Pillai, P. P.*, Accelerated Reduction of 4-Nitrophenol: Bridging Interaction Outplays Reducing Power in the Model Nanoparticle Catalyzed Reaction *J. Phys. Chem. C*, **2020**, *124*, 35, 19157 – 19165. (* as co-corresponding author).
13. **Rao, A.**, Roy, S., Pillai, P. P.*, Temporal Changes in Interparticle Interactions Drive the Formation of Transiently Stable Nanoparticle Precipitates *Langmuir*, **2021**, *37*, 1843 – 1849.
14. Chakraborty, I. N., Roy, P., **Rao, A.**, Devatha, G., Roy, S., Pillai, P. P.*, The Unconventional Role of Surface Ligands in Dictating the Light Harvesting Properties of Quantum Dots *J. Mater. Chem. A*, **2021**, *9*, 7422 – 7457.
15. Roy, S., Adury, V. S. S., **Rao, A.**, Roy, S., Mukherjee, A., Pillai, P. P.*, Electrostatically Directed Long-Range Self-Assembly of Nucleotides with Cationic Nanoparticles To Form Multifunctional Bioplasmonic Networks *Angew. Chem. Int. Ed.*, **2022**, *61*, e202203924.
16. **Rao, A.***, Roy, S., Jain, V., Pillai, P. P.*, Nanoparticle Self-Assembly: From Design Principles to Complex Matter to Functional Materials *ACS Appl. Mater. Interfaces*, **2022**, *15*, 25248 – 25274. (* as co-corresponding author).
17. Kashyap, R. K., Dwivedi, I., Roy, S., Roy, S., **Rao, A.**, Subramaniam, C., Pillai, P. P.*, Insights into the Utilization and Quantification of Thermoplasmonic Properties in Gold Nanorod Arrays *Chem. Mater.*, **2022**, *34*, 7369 – 7378.
18. Eatson, J. L., Gordon, J. R., Cegielski, P., Giesecke, A. L., Suckow, S., **Rao, A.**, Silvestre, O. F., Liz-Marzán, L. M., Horozov, T. S. Buzza, D. M. A.* Capillary Assembly of Anisotropic Particles at Cylindrical Fluid–Fluid Interfaces *Langmuir*, **2023**, *39*, 6006 – 6017.
19. Silvestre, O. F.#, **Rao, A.**#, Liz-Marzán, L. M.*, Self-Assembled Colloidal Gold Nanoparticles as Substrates for Plasmon Enhanced Fluorescence *Eur. J. Mater.*, **2023**, *1*, 2202676. (# equal

contribution).

20. Kruse, J., **Rao, A.**, Sánchez-Iglesias, A., Montaña-Priede, J. L., Ibarra, A. I., Lopez, E., Seifert, A., Arbe, A., Grzelczak, M.*, Temperature-modulated Reversible Clustering of Gold Nanorods Driven by Small Surface Ligands *Chem. - Eur. J.*, **2023**, e202302793.
21. Jumbo-Nogales, A., **Rao, A.**, Olejniczak, A., Grzelczak, M., Rakovich, Y.*, Unveiling the Synergy of Coupled Gold Nanoparticles and J-Aggregates in Plexcitonic Systems for Enhanced Photochemical Applications, *Nanomaterials*, **2024**, 14, 35.
22. **Rao, A.***, Grzelczak, M.*, Revisiting El-Sayed Synthesis: Bayesian Optimization for Revealing New Insights during the Growth of Gold Nanorods, *Chem. Mater.*, **2024**, 36, 2577 – 2587. (* as co-corresponding author).
23. **Rao, A.***, Sánchez-Iglesias, A., Grzelczak, M.*, Choreographing Oscillatory Hydrodynamics with DNA-Coated Gold Nanoparticles. *J. Am. Chem. Soc.*, **2024**, 146, 18236 – 18240. (* as co-corresponding author).
24. Montaña-Priede, J. L., **Rao, A.**, Sánchez-Iglesias, A., Accelerated Design of Gold Nanoparticles with Enhanced Plasmonic Performance, *ChemRxiv*, DOI: 10.26434/chemrxiv-2025-fckd0.
25. Naik, A. M., Sánchez-Iglesias, A., Montaña-Priede, J. L., D'souza, N. M., Sancho-Parramon, J., Mezzasalma, S. A., **Rao, A.***, Grzelczak, M.*, Size Effect on Photothermal Heating Ability of Gold Bipyramids. *Adv. Optical Mater.*, **2025**, e01006 (* as co-corresponding author).

Manuscripts in preparation.....

26. **Rao, A.**#, Renero-Lecuna, C.#, Quintanilla, M., Bergamini, L., Zabala, N., Kuttner, C., Aizpurua, J., Liz-Marzán, L. M.*, Higher absorption defies lower efficiency: Effect of Particle Size on Photothermal Heating by Gold Nanorods (# equal contribution).
27. Jumbo-Nogales, A., **Rao, A.**, Olejniczak, A., Grzelczak, M.*, Rakovich, Y.*, pH-Switchable Strongly Coupled Plexitonic Systems with Gold Bipyramids and J-Aggregates.
28. Naik, A. M., Sánchez-Iglesias, A., Montaña-Priede, J. L., **Rao, A.**, Grzelczak, M.*, Effect of Light on the Overgrowth of Silver on Gold Bipyramids.
29. **Rao, A.***, Sánchez-Iglesias, A., Grzelczak, M.*, Closed-loop Optimization and Synthesis of Plasmonic Nanoparticles. (* as co-corresponding author).

EXPERIMENTAL TECHNIQUES AND SKILLS

I am experienced in the following areas:-

- **Design and synthesis of nanoparticles (NPs):** Expertise in the synthesis of plasmonic, semiconductor, and dielectric NPs, along with their surface modification.
- **Multicomponent NP assemblies:** Proficient in leveraging electrostatic and H-Bonding interactions to produce high-yield multicomponent NP assemblies.
- **Photothermal therapy:** Development of efficient NP systems tailored for applications in photothermal therapy.
- **Responsive NP self-assemblies:** Skilled in designing both equilibrium and non-equilibrium NP self-assemblies, which offer promising applications for next-generation sensor, and catalyst technologies.
- **Stabilization of unstable entities:** Specializing in the use of ligand chemistry to stabilize inherently unstable nanomaterials.
- **Design of identification assays:** Creation of innovative assays capable of demonstrating emergent properties, such as selectivity, that are absent in the individual components.
- **Catalyst design for AuNP-catalyzed reactions:** Expertise in controlling reaction outcomes to develop efficient catalysts, including those used for the regeneration of NADH.
- **Data analysis and automation:** Experienced in Python programming for data analysis, Bayesian optimization, machine learning, and hardware programming to automate experiments.

CONFERENCES, SCHOOLS AND WORKSHOPS ATTENDED

- Presented a poster at **GRC Plasmonically Powered Processes**, Ventura, titled '*Navigating Complexity: Designing Transient and Oscillating Systems with Nanoparticles*' from June 8 - 13, 2025.
- Gave a talk titled '*From Human Hands to AI and Robotics: Building Nanoparticle-Based Complex Systems*' at **SoftComp/EUSMI Annual Meeting 2025** at **Venice-Mestre, Italy** on May 19 – 22nd, 2025.
- Presented a poster at **Dissipative Processes in Molecular Systems (DpMS)** Padova, titled '*Navigating Complexity: Designing Transient and Oscillating Systems with Nanoparticles*' from June 17 - 21, 2024.
- Gave a talk titled '*Using AI for Accelerating Nanoparticle Synthesis*' at **4th Quantum Breakfast at CFM, Donostia-San Sebastian, Spain** in 26 April 2024.
- Presented a poster at **Translight Workshop between Bordeaux-San Sebastian on Quantum Nanophotonics** titled '*Machine Learning for the Synthesis of Gold Nanorods*' from December 18 - 19, 2023.
- Gave a talk titled '*Machine Learning for Nanoparticle Synthesis: Transitioning From 'How to Make Things' to 'How Things Are Made'*' in **AI Photonics 2023** at **Donostia-San Sebastian, Spain** in September 2023.
- Presented a poster at **Gordon Research Conference on Self-Assembly and Supramolecular Chemistry** titled "*Regulating Nanoscale Forces to Control Macroscale Functions*" from May 19 - 24, 2019.
- Presented a poster as well as a short 90-second talk at **Alexander von Humboldt (AvH) Kolleg** titled "*Regulating Nanoscale Forces to Control Macroscale Functions*" from January 31 to February 2, 2019 at **Kashid**.
- Attended **Indo-US Workshop on Soft Matter (IUWSM-2018)** at **IIT Roorkee** from December 9 - 11, 2018.
- Gave a talk titled "*Emergence of Selectivity in Inherently Nonselective Gold Nanoparticles by Controlling Interparticle Interactions*" in **Chemsymphoria (in-house Symposium)** at **IISER Pune** in July 2018.
- Attended the Conference **Innovations in Frontier Chemistry (IFC)** 2018 held at **IISER Pune** from May 8th – 9th.
- Attended **DST School on Nanoscience and Nanotechnology** at **CeNS Bengaluru** from 23rd October to 3rd November 2017.
- Presented a poster titled "*Regulating Interparticle Forces Reveal Phenomenon of Controlled Aggregation in Charged Nanoparticles*" in **International Conference on Nanoscience and Technology (ICONSAT)** held at **IISER-Pune** in March 2016.

POSITIONS OF RESPONSIBILITY

TEACHING AND ADVISING.....

- I am currently a **teaching assistant (T.A.)** for **Modern Optics** course offered to postgraduates at University of Fribourg.
- **T.A. for Advanced Physical Chemistry Lab:** Assisted in the design of experiments and led weekly lab sessions for a class of approximately 25 Fourth-Year undergraduate students.
- **T.A. for Quantum Chemistry:** Conducted weekly tutorial sessions and office hours, and graded assignments for a class of around 50 Third-Year undergraduate students.
- **T.A. for Chemical Principles I:** Planned and led discussions during tutorial sessions and remedial classes, helped design assignments, conducted weekly office hours, and graded papers and exams for a class of approximately 200 First-Year undergraduate students.
- **Mentorship of Fifth-Year Master's Thesis Students:** Assisted in mentoring four students on their thesis projects: Govind Sarang, Ajesh T. Rajesh, Sarah Peneena K. J., and Shana Shirin V.
- **Mentorship of Summer Project Students:** Guided five summer project students from various institutes in India: Mahima Unnikrishnan, Preethi Susan Mathew, Jewel Ann Maria, Sarah Peneena K. J., and Shana Shirin V.

- **Mentorship of PhD Students:** Supported and mentored PhD students Sumit Roy and Alba Jumbo Nogales during their doctoral research.
- **Mentor for Sci-ROI@India's ReAct Program (2022):** Selected as one of nine mentors for the program's first cohort, where I remotely mentored two undergraduate students from India on a project studying photothermal heating with graphite for water desalination.

ADMINISTRATIVE EXPERIENCE.....

- Actively participated in the organization of ICONSAT (*International Conference on Nanoscience and Nanotechnology*) - with more than ~ 400 participants, at IISER Pune in 2016.
- Was part of the *Social Media Team* associated with *India March for Science, Pune Chapter* in 2017. Here, actively participated in popularizing the march across different colleges and places in Pune.
- As Joint Secretary of the Chemphilic Club, at IISER Pune, helped in organizing an in-house Symposium (ChemSymphoria), New Year's Party, and Departmental Excursion.

AWARDS AND ACHIEVEMENTS

- **July 2016** - Cleared CSIR-NET (JRF) with AIR - 77. This is a national-level exam conducted by the Indian government.
- **2019** - Got CSIR Travel Grant to present at the 2019 Gordon Research Conference (GRC) on Self Assembly and Supramolecular Chemistry, in Switzerland.
- **2019** - Got Infosys Travel Grant to register and present at the 2019 GRC on Self Assembly and Supramolecular Chemistry, in Switzerland.
- **2021** - Got the prestigious Juan de la Cierva fellowship to pursue research at Centro de Física de Materiales (CFM), Spain.